

Designing Sound Synthesis Interfaces for Head-mounted Augmented Reality

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ABSTRACT

We report on designing a sound synthesis interface for a head-mounted augmented reality environment. The increased accessibility of augmented reality (AR) devices has incentivised the exploration of sound applications for music performance in computer music and other relevant communities. However, interaction affordances vary based on the specific AR device and thus implies different design considerations. In this poster, we present different interface prototypes for a frequency modulation synthesis system in the Microsoft HoloLens 2 and report our insights during the process of their developments.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed/augmented reality; Applied computing—Arts and humanities—Sound and music computing

1 INTRODUCTION

The increased accessibility of augmented reality (AR) devices has presented a new digital space for music experience and performance. In the field of musical interface design, related works have presented different interfaces for sound synthesis based on the specific AR device that is used. For example, the AuSynthAR [6], is a low-cost sound synthesiser that is a combination of software and physical tokens using a mobile device. More recent works use advanced AR devices such as head-mounted displays (HMDs), which support a more natural user interface for music production and experience. In [3], a head-mounted augmented reality (HMAR) compositional platform is created which allows the user to use eye-gaze or body movements during electronic music production. In contrast, [11] considers free-hand gesturing to manipulate sound components for an augmented sonic experience. Other relevant works such as [9] presented a modular synthesis through Oculus hand-held controllers in virtual reality. Berthaut [4] provided a summary of 3D interaction techniques for musical expression, where he discussed different interaction techniques as well as how these techniques could be used for immersive music interfaces.

With these examples in mind, our work is motivated by two questions: 1) What is a genuine interface for music expression that suits a HMAR environment? 2) How would the interaction affect music making? In this poster, we present three different interface prototypes that are specifically developed for a frequency modulation synthesis system in HMAR. We describe the details of each interface, reflect on the experience of their development and discuss future works.

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2 SYSTEM DESIGN

Our goal is to design a suitable interface for sound synthesis within the HMAR environment. The interface should provide a natural sound synthesis experience through the affordance and user experience of HMD devices. Our choice of AR device is the Microsoft HoloLens 2 (HL2) [7] which supports hand tracking and has the advantage of flexibility that is similar to how we would interact with a physical synthesiser such as turning knobs. The HL2 offers a range of free-hand interaction options including NearinteractionTouchable, NearinteractionGrabbable, far pointer interactions, etc. [2], allowing the user to interact with virtual objects as one would with physical objects.

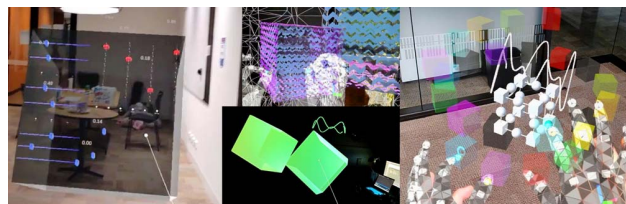


Figure 1: Three interface prototypes seen through the HL2: direct control, ambiguous control and finger interaction (left to right).

2.1 Frequency Modulation Synthesis

We selected a simple frequency modulation synthesiser as the sonic choice for our user interface. The basic components of this synthesis routine are: two oscillators and their properties, a mixer for these two oscillators, an ADSR (attack-decay-sustain-release) envelope, LFO (low-frequency oscillator) and filter(s).

Sound generation was implemented with the Disunity library [10] in Unity. Disunity provides a sound synthesis primitives such as oscillators, envelopes, mixer, etc. These synthesis components can be assembled by feeding into each other's input in sequence and then output to an audio sink in Unity.

2.2 Prototypes

2.2.1 Design Process

The design of our interface started with the inspiration of modelling an interface from physical sound synthesisers, that is, a virtual panel containing sliders and buttons mapping to frequency modulation synthesis parameters. Soon we found that using hand gestures to control these parameters requires accurate manipulation that costs an amount of time due to the nature of hologram instability in the HMAR environment [1]. This affects the process of making sound in real-time while tuning sound parameters.

To reduce this side-effect, our design shifted to an “ambiguous” user interface which transforms the direct control of sliders and buttons to a more simple and easy interaction through sound cubes. We also considered embodied interaction which maps between sound parameters and the way that the user naturally interact with virtual objects in the HMAR environment. In other words, the functionality

of the interface remains the same but presents a different way of controlling sound parameters. By experimenting with this interface, we found the ease of controlling and changing sound properties was increased by aggregating several sound parameters as a whole, but also noticed that the interface lacks the diversity or richness as a musical instrument, particularly in the situation of music performance.

Therefore, in our final design, we took the idea of having one interaction that maps to the control of several pre-fixed sound parameters, and created an interface that is composed of individual sound cubes with each representing a musical note. The interaction also shifts to finger interaction instead to allow multiple notes to be played at the same time. We give details of each above interface in the following subsections.

2.2.2 Development

The development of user interfaces was realised in Unity version 2019.4.31f1 with Microsoft's Mixed Reality Toolkit (MRTK) [8]. The interface is composed of interactable 3D objects scripted by MRTK and Disunity as game objects in Unity.

2.2.3 Prototype One: Direct Control

As shown in Figure 2, this interface is modelled from the design of a physical sound synthesiser that has knobs, sliders and buttons. Here, the 3d cube acts as a panel with sliders and buttons for tuning sound properties. At the left side of the interface, it offers different sound wave type options of two oscillators, and provides sliders to adjust their sound gains and frequencies for sound modulation. While at the right side, an ADSR envelope (top) and a filter with different sound properties' options are presented for tuning through sliders or buttons.

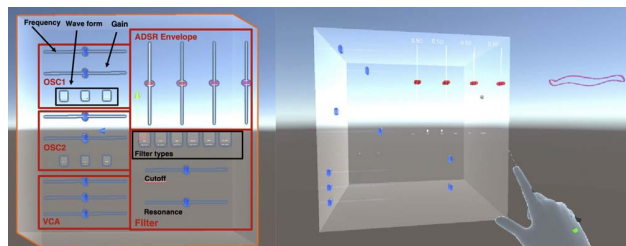


Figure 2: The interface that contains sliders and buttons to tune FM synthesiser sound parameters (left). Attempting to adjust sliders on the interface in Unity, with the audio visualisation (right).

2.2.4 Prototype Two: Ambiguous Control

In contrast to the above interface having a cube as a control panel, this interface (Figure 3) is composed of two cubes with each representing one oscillator (one modulates the other). All control of sound parameters is embodied interaction allowing the user to freely walk around and keep sound synthesising. This is achieved by interacting with these sound cubes, including manipulating horizontally (LFO), vertically (mapping frequency control) and increasing/shrinking their sizes (volume). The option for the filter is hidden from the interface with pre-defined settings.

2.2.5 Prototype Three: Finger Composing

Our final design is an interface containing a series of small sound cubes, arranged in a cylinder surface in an anticlockwise direction like a MIDI keyboard (Figure 4). Similar to the second prototype, each sound cube is a musical note made by FM synthesis with the same pre-defined sound parameters. The design of the cylinder surface offers ease of interaction, allowing the user to use fingers to play chords and providing more musical composition options.

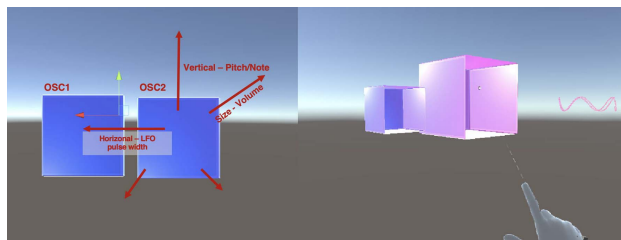


Figure 3: The interface by two cubes having sound embedded (left). Changing pitch through moving a cube vertically in Unity (right).

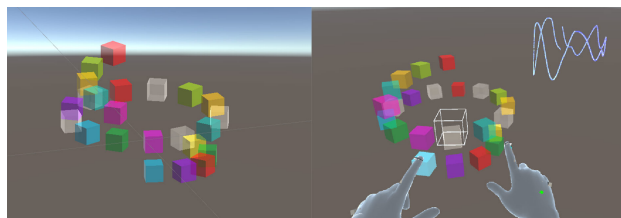


Figure 4: Music notes including tones and semitones arranged in a cylinder surface (left). Playing chords in Unity (right).

Additional visuals are attached to these sound cubes to distinguish tones and semitones on this interface by different colours.

3 REFLECTION AND FUTURE DIRECTION

Through developing three user interfaces, we report our experience of using them and discuss insights obtained for designing a new musical interface of sound synthesis for the HMAR environment.

We first found that the experience of controlling sound parameters in HMAR is different from how we would interact with a physical sound synthesiser. The instability of holograms as the user interface presents a difficulty for the user to perform an accurate hand interaction such as sliding a knob. While trying out the first interface, the author, who has nearly two years of experience with HL2, still found it difficult and felt irritated about spending a fair amount of time adjusting these sliders. However, having reduced these subtle interactions and not offering all sound parameters' options in the second prototype, we increased ease-of-use and could quickly obtain a sonic response "ambiguous" control interactions. These experiences suggest that the design of the sound synthesis interface could avoid fine or subtle tuning and adapt to the HMAR environment given its 3D interaction affordance.

The above discussion also leads to another core idea for the design of a new musical interface. In [5], Dobrian and Koppelman distinguished concepts of "design for sound control" and "design for sound expression", suggesting a proper new musical interface should not only satisfy the basic functionality of sound control, but also possess artistic qualities such as "being expressive" or "virtuosity" considering its performing purpose. That said, it can be seen that our prototypes' development slowly moved from a purely functional interface to an ambiguous or even an artistic interface, although they are still far from a sophisticated musical instrument. These insights remind us of the mapping between input control and sonic feedback for designing musical expression in a broader context.

With these insights, our future work will focus on refining developed prototypes based on our reflections and conducting systematic user studies through music performances to examine their effectiveness as an interface of musical expression for the HMAR environment.

4 CONCLUSION

We presented three user interfaces for sound synthesis in an HMAR environment using HL2. Through their design and development, we reflected on the design experience of mapping between input interaction and sonic feedback. With these empirical results we provide useful insights into the new musical interface design for HMAR.

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